

Lecture 6: Vector space model and probabilistic ranking

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Today's lecture

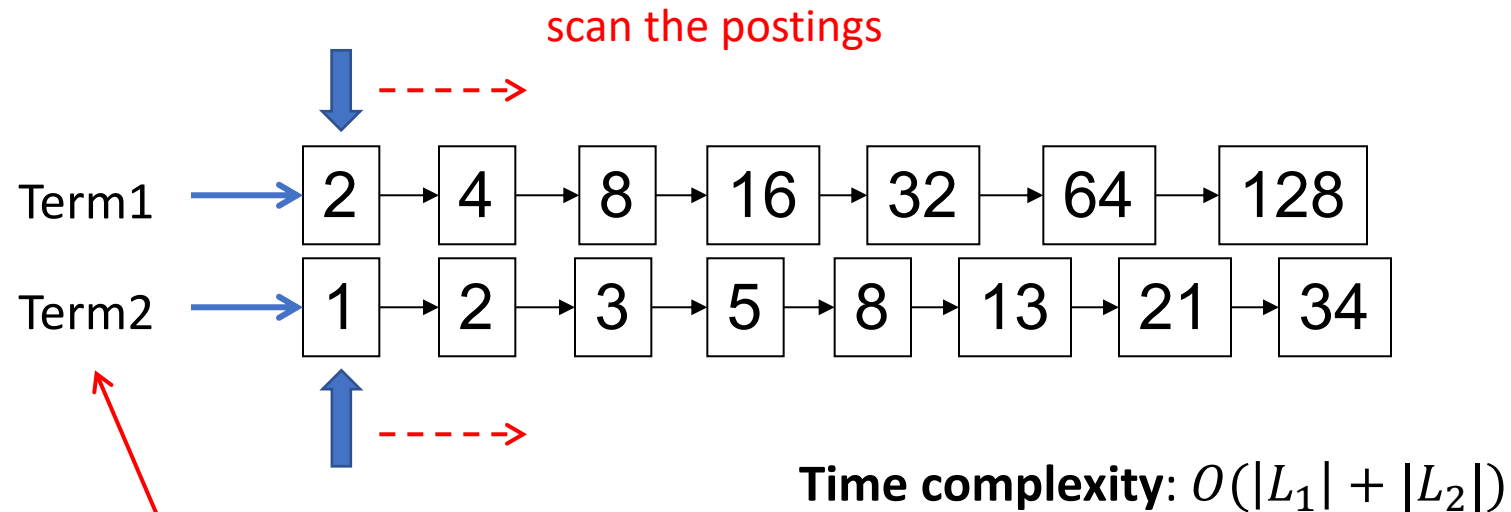
- Boolean model
- Vector space model
- Probabilistic ranking principle

Boolean model

Search with Boolean query

- Boolean query
 - E.g., “obama” AND “healthcare” NOT “news”
- Procedures
 - Lookup query term in the dictionary
 - Retrieve the posting lists
 - Operation
 - AND: intersect the posting lists
 - OR: union the posting list
 - NOT: diff the posting list

Recap: AND operation



Trick for speed-up: when performing multi-way join, starts from lowest frequency term to highest frequency ones

Deficiency of Boolean model

- The query is unlikely precise
 - “Over-constrained” query (terms are too specific): no relevant documents can be found
 - “Under-constrained” query (terms are too general): over delivery
 - It is hard to find the right position between these two extremes (hard for users to specify constraints)
- Even if it is accurate
 - **Not** all users would like to use such queries
 - **Not** all relevant documents are **equally** important
 - No one would go through all the matched results
- Relevance is a matter of degree!

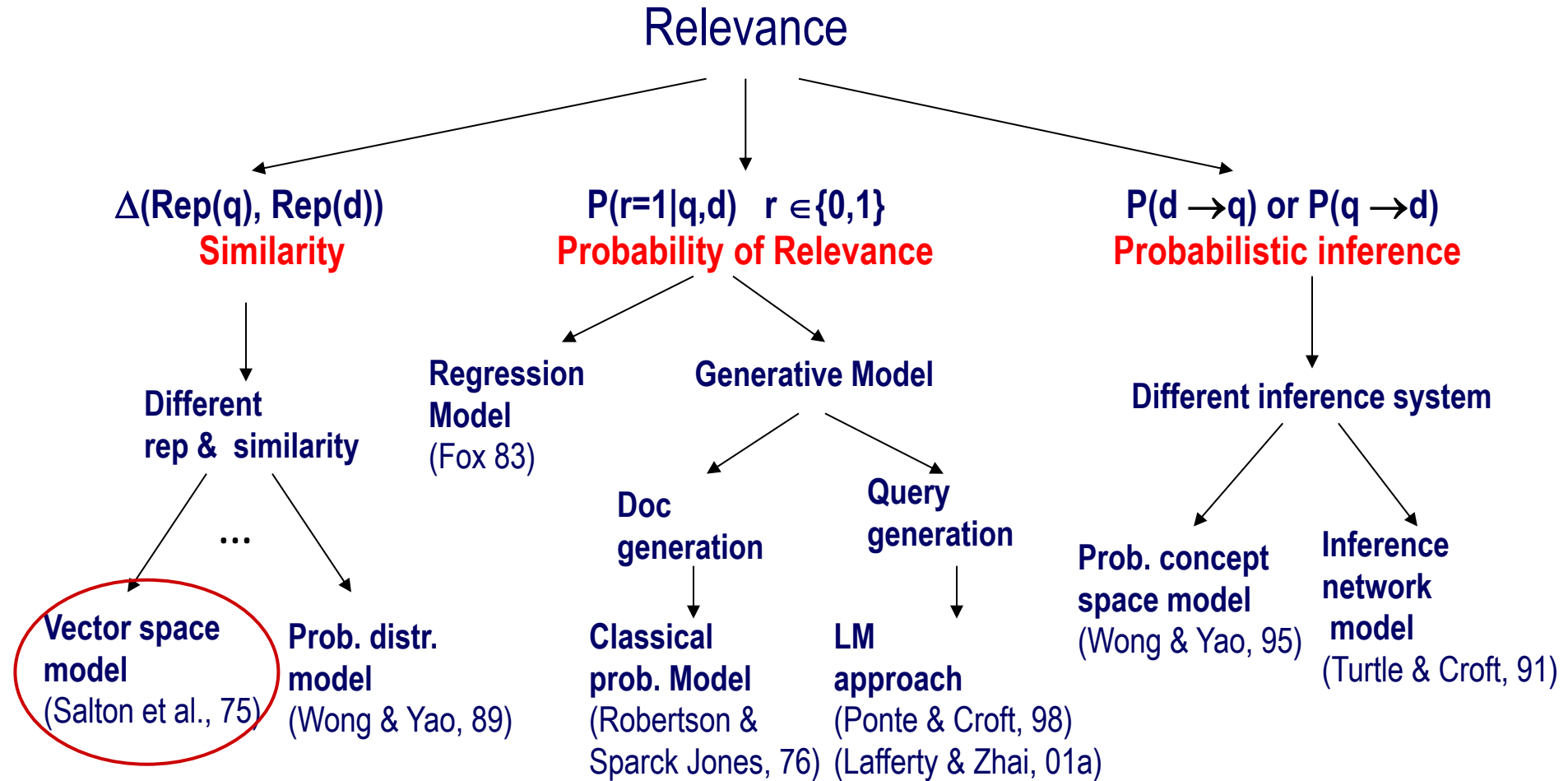
Ranking is often preferred

- Relevance is a matter of degree
 - Easier for users to find appropriate queries
- A user can stop browsing anywhere, so the boundary is controlled by the user
 - Users prefer coverage would view more items
 - Users prefer precision would view only a few
- Theoretical justification: Probability Ranking Principle

Retrieval procedure in modern IR

- Boolean model provides all the ranking candidates
 - Locate documents satisfying Boolean condition
 - E.g., “obama healthcare” -> “obama” OR “healthcare”
- Rank candidates by relevance
 - Important: the notion of relevance
- Efficiency consideration
 - Top-k retrieval

Notion of relevance



Intuitive understanding of relevance

- Fill in magic numbers to describe the relation between documents and words

	information	retrieval	retrieved	is	helpful	for	you	everyone
Doc1	1	1	0	1	1	1	0	1
Doc2	1	0	1	1	1	1	1	0
Query	1	1	0	0	0	0	0	0



*E.g., 0/1 for Boolean models,
probabilities for probabilistic models*

Some notations

- Vocabulary $V = \{w_1, w_2, \dots, w_N\}$ of language
- Query $q = t_1, \dots, t_m$, where $t_i \in V$
- Document $d_i = t_{i1}, \dots, t_{in}$, where $t_{ij} \in V$
- Collection $C = \{d_1, \dots, d_k\}$
- $\text{Rel}(q, d)$: relevance of doc d to query q
- $\text{Rep}(d)$: representation of document d
- $\text{Rep}(q)$: representation of query q

Vector space model

Relevance = Similarity

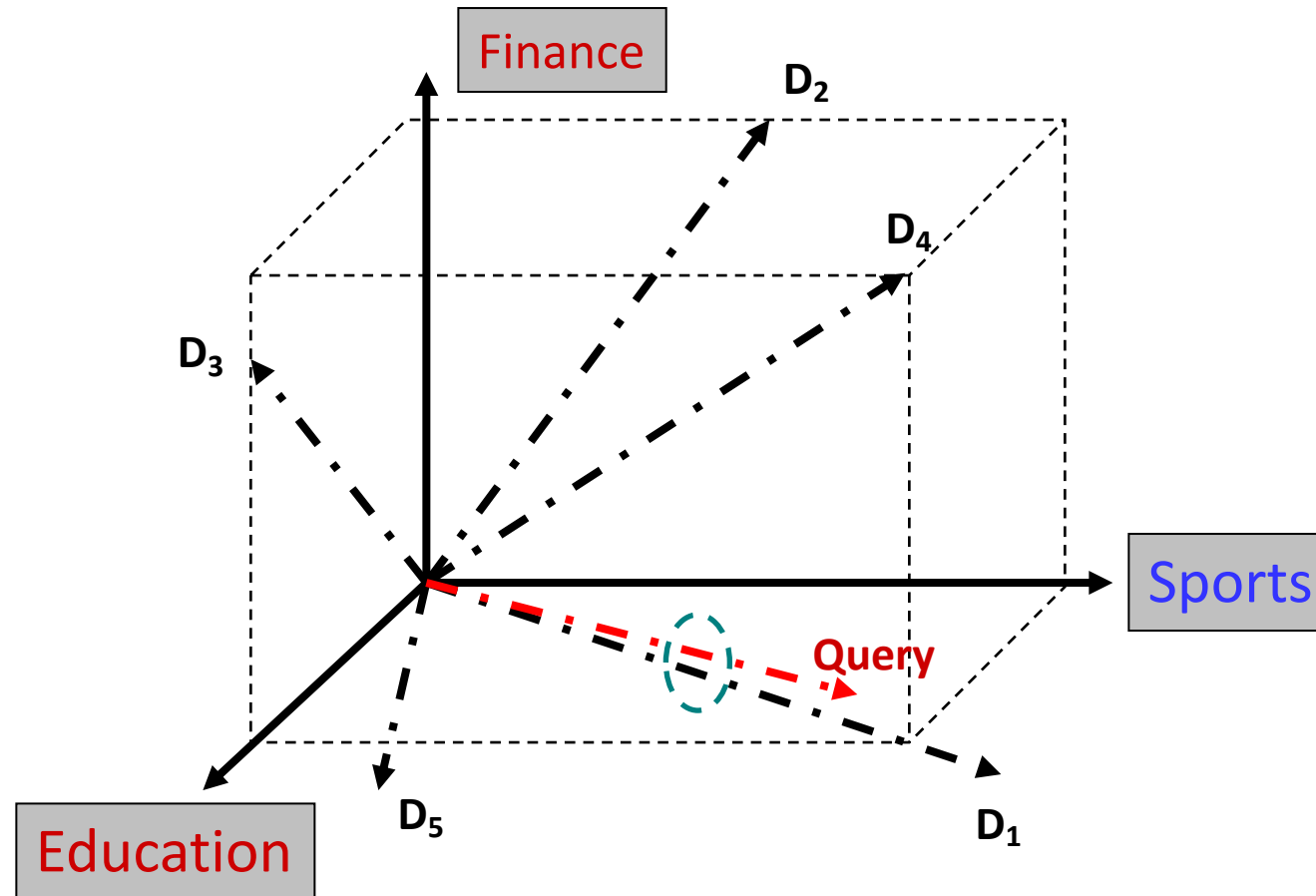
- Assumptions
 - Query and documents are represented in the same form
 - A query can be regarded as a “document”
 - $\text{Relevance}(d,q) \propto \text{similarity}(d,q)$
- $R(q) = \{d \in C \mid \text{rel}(d,q) > \theta\}$, $\text{rel}(q,d) = \Delta(\text{Rep}(q), \text{Rep}(d))$
- Key issues
 - How to represent query/document?
 - How to define the similarity measure $\Delta(x,y)$?

Vector space model

- Represent both document and query by concept vectors
 - Each concept defines one dimension
 - K concepts define a high-dimensional space
 - Element of vector corresponds to concept weight
 - E.g., $d=(x_1, \dots, x_k)$, x_i is “importance” of concept i
- Measure relevance
 - Distance between the query vector and document vector in this concept space

VS Model: an illustration

- Which document is closer to the query?



More details

- How to define/select the “basic concept”
 - Concepts are assumed to be orthogonal
- How to assign weights
 - Weight in query indicates importance of the concept to the user’s information need
 - Weight in doc indicates how well the concept characterizes the doc
- How to define the similarity/distance measure

What is a good “basic concept”?

- Orthogonal
 - Linearly independent basis vectors
 - “Non-overlapping” in meaning
 - No ambiguity
- Weights can be assigned automatically and accurately
- Existing solutions
 - Terms or N-grams, i.e., bag-of-words
 - Topics, i.e., topic model

How to assign weights?

- Important!
- Why?
 - Query side: not all terms are equally important
 - Document side: some terms carry more information about the content
- How?
 - Two basic heuristics
 - TF (Term Frequency) = Within-doc-frequency
 - IDF (Inverse Document Frequency)

TF weighting

- Idea: a term is more important if it occurs more frequently in a document
- TF Formulas
 - Let $f(t, d)$ be the frequency count of term t in doc d
 - Raw TF: $tf(t, d) = f(t, d)$

TF normalization

- Two views of document length
 - A doc is long because it is verbose
 - A doc is long because it has more content
- Raw TF is inaccurate
 - Document length variation
 - “Repeated occurrences” are less informative than the “first occurrence”
 - Relevance does not increase proportionally with number of term occurrence
- Generally penalize long doc, but avoid over-penalizing
 - Pivoted length normalization

Query: *iphone 16*

D1: iPhone 16 receives pre-orders on September.

D2: iPhone 16 has six color options.

D3: iPhone 16 has six color options.

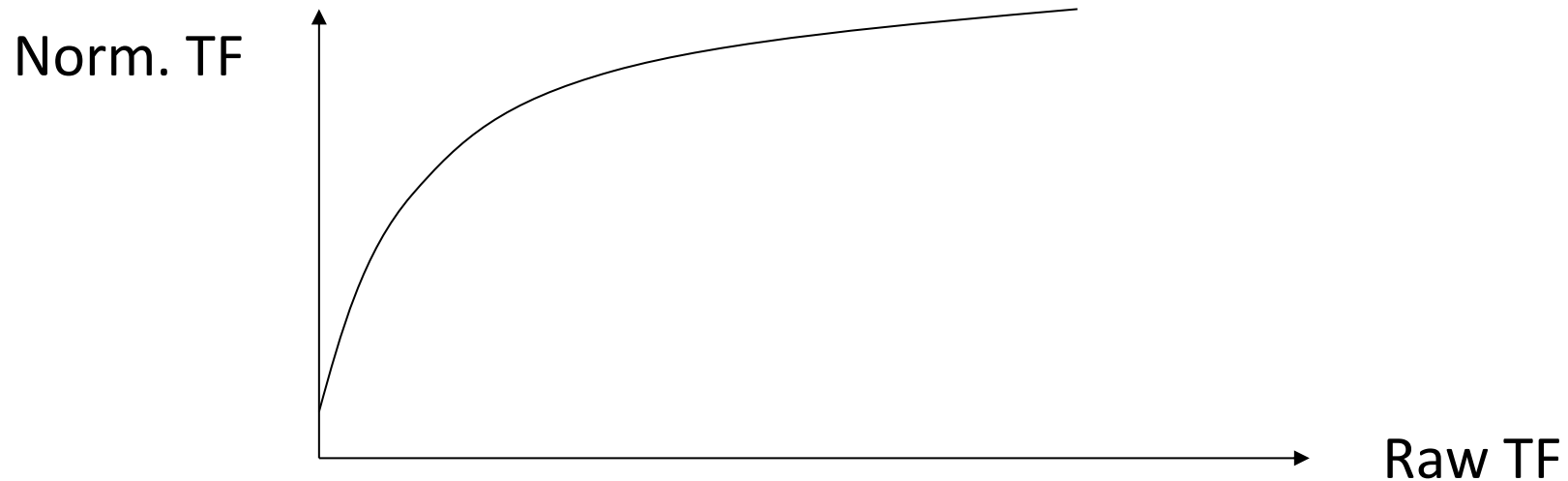
iPhone 16 has six color options.

iPhone 16 has six color options.

TF normalization

- Sublinear TF scaling

- $tf(t, d) = \begin{cases} 1 + \log f(t, d), & \text{if } f(t, d) > 0 \\ 0, & \text{otherwise} \end{cases}$

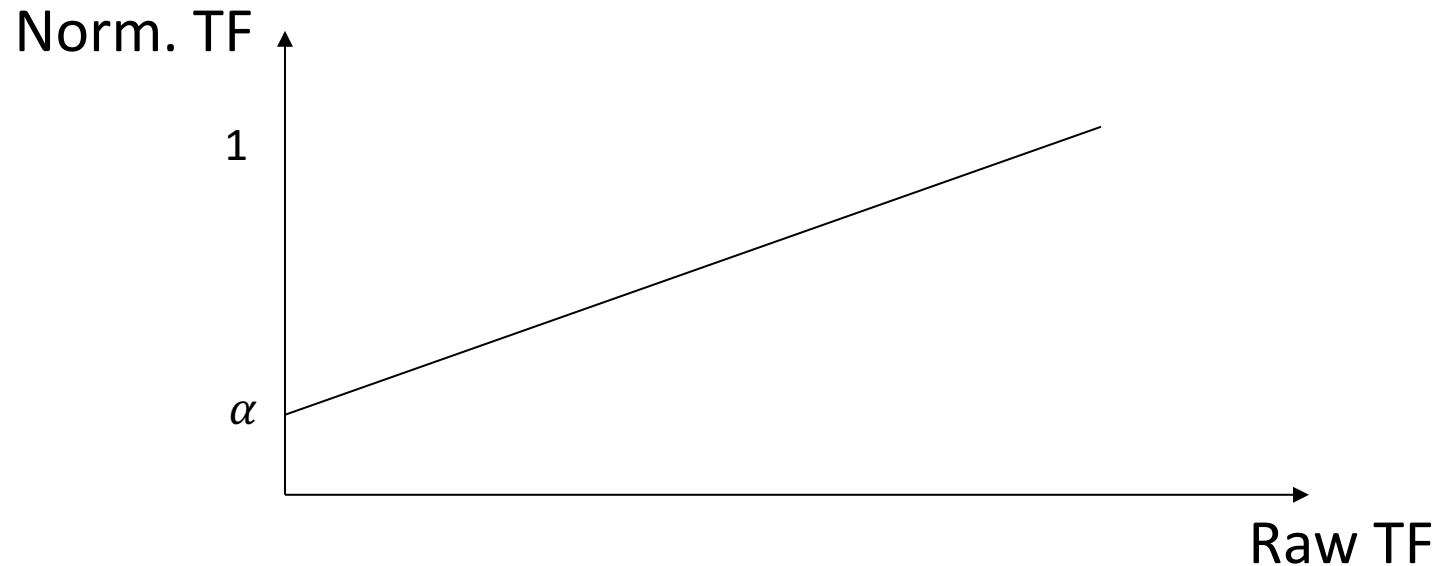


TF normalization

- Maximum TF scaling

- $tf(t, d) = \alpha + (1 - \alpha) \frac{f(t, d)}{\max_t f(t, d)}$

- Normalize by the most frequent word in this doc



Document frequency

- Idea: a term is more discriminative if it occurs only in fewer documents

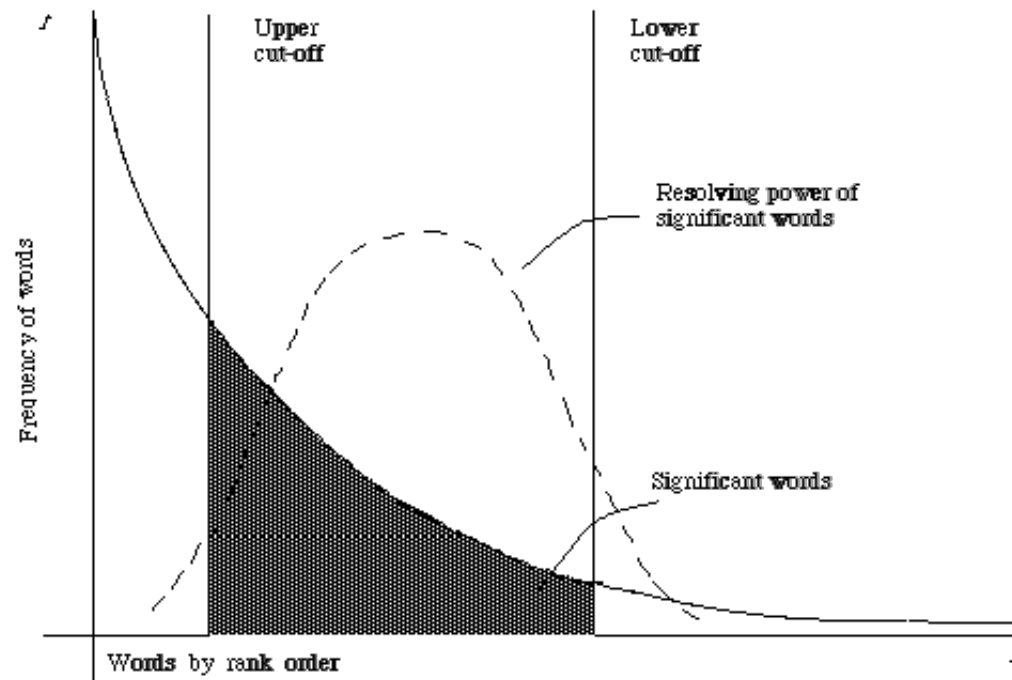


Figure 2.1. A plot of the hyperbolic curve relating f , the frequency of occurrence and r , the rank order (Adapted from Schultz⁴⁴ page 120)

IDF weighting

- Solution

- Assign higher weights to the rare terms

- Formula

- $IDF(t) = 1 + \log\left(\frac{N}{df(t)}\right)$

Non-linear scaling

Total number of docs in collection

Number of docs containing term t

- A corpus-specific property

- Independent of a single document

Quiz

- If we remove a document from the corpus, how would it affect a term's IDF?

Why document frequency

- How about total term frequency?

- $ttf(t) = \sum_d f(t, d)$

Table 1. Example total term frequency v.s. document frequency in Reuters-RCV1 collection.

Word	ttf	df
try	10422	8760
insurance	10440	3997

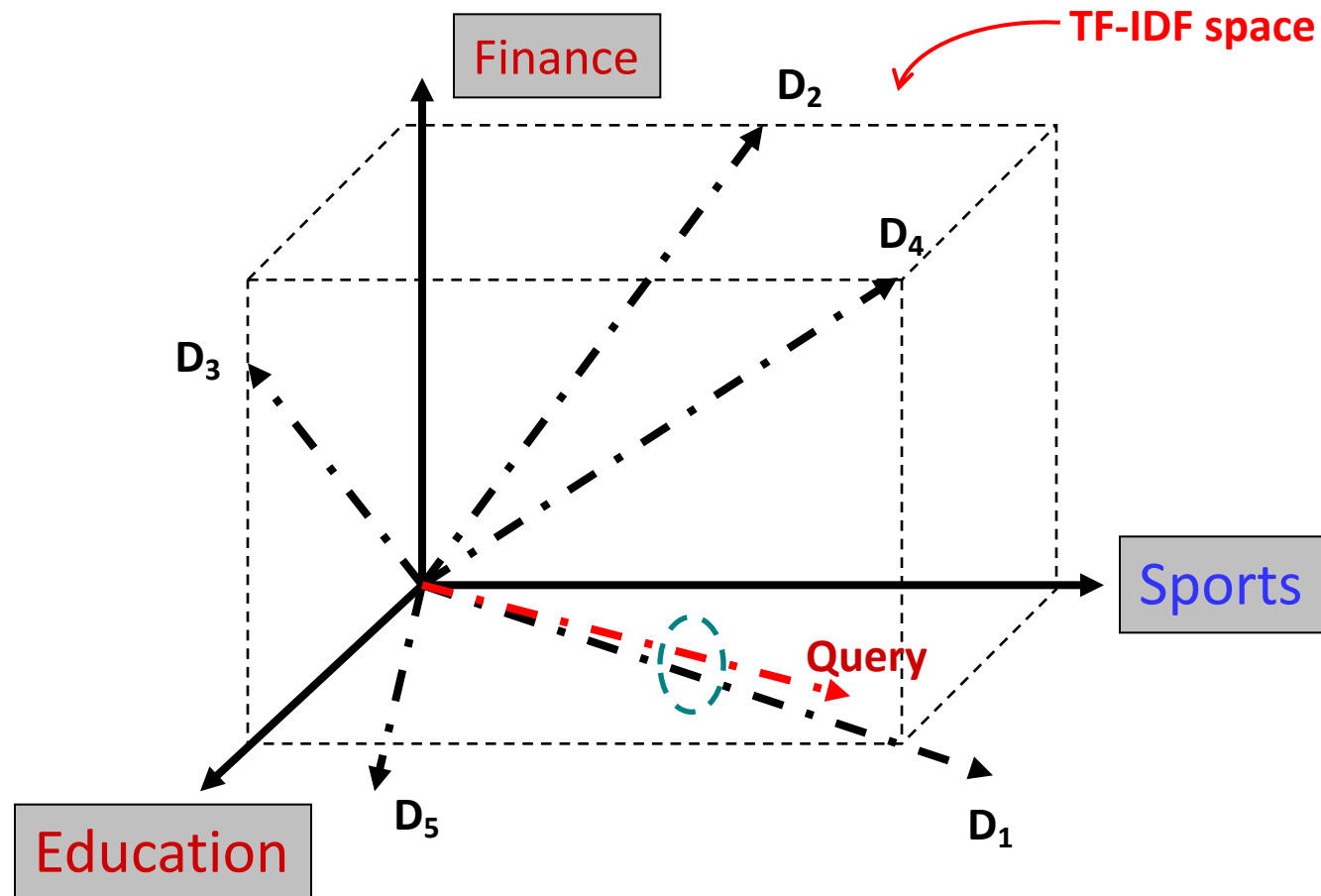
- Cannot recognize words frequently occurring in a subset of documents

TF-IDF weighting

- Combining TF and IDF
 - Common in doc $\rightarrow$ high tf $\rightarrow$ high weight
 - Rare in collection $\rightarrow$ high idf $\rightarrow$ high weight
 - $w(t, d) = TF(t, d) \times IDF(t)$
- Most well-known term weighting schema in IR! (G Salton et al. 1983)

How to define a good similarity measure?

- Euclidean distance?



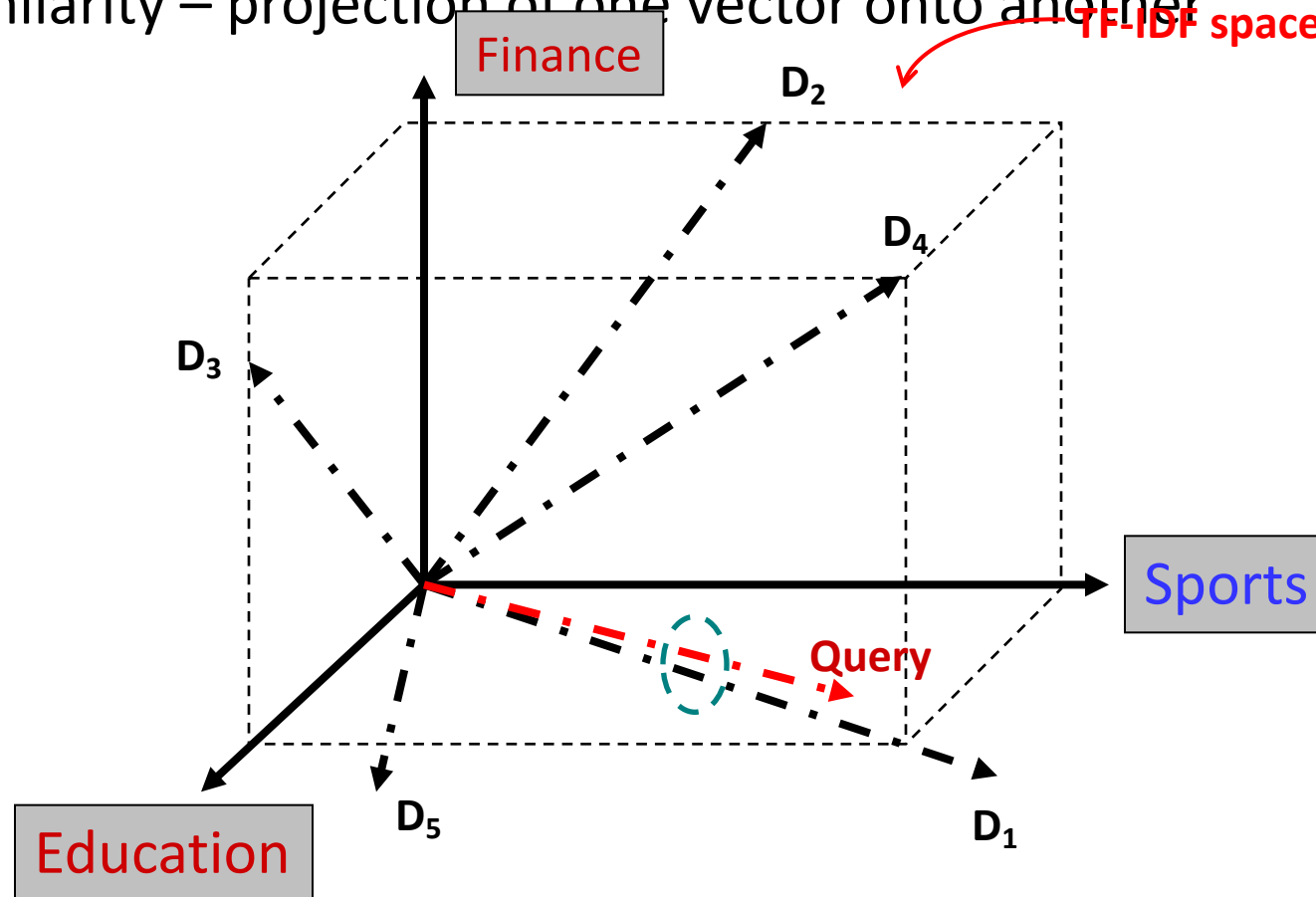
How to define a good similarity measure?

- Euclidean distance

- $dist(q, d) = \sqrt{\sum_{t \in V} [tf(t, q)idf(t) - tf(t, d)idf(t)]^2}$
- Longer documents will be penalized by the extra words
- We care more about how these two vectors are overlapped

From distance to angle

- Angle: how vectors are overlapped
 - Cosine similarity – projection of one vector onto another



Cosine similarity: from distance to angle

- Angle: how vectors are overlapped
 - Cosine similarity – projection of one vector onto another

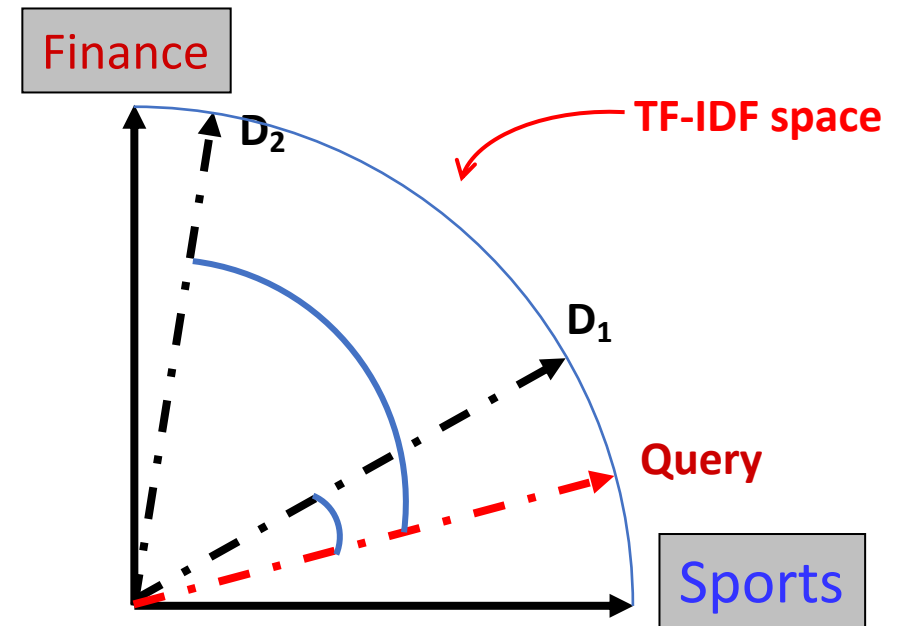
- Angle between two vectors

$$\text{cosine}(V_q, V_d) = \frac{V_q \times V_d}{|V_q|_2 \times |V_d|_2} = \frac{V_q}{|V_q|_2} \times \frac{V_d}{|V_d|_2}$$

- Document length normalized

TF-IDF vector

Unit vector



Fast computation of cosine in retrieval

- $\text{cosine}(V_q, V_d) = V_q \times \frac{V_d}{|V_d|_2}$
 - $|V_q|_2$ is the same for all candidate docs
 - Normalization of V_d can be done in indexing time
 - Only count $t \in q \cap d$
 - Score accumulator for each query term when intersecting postings from inverted index

Advantages of VS Model

- Empirically effective! (Top TREC performance)
- Intuitive
- Easy to implement
- Well-studied/Most evaluated
- The Smart system
 - Developed at Cornell: 1960-1999
 - Still widely used
- Warning: Many variants of TF-IDF!

Disadvantages of VS Model

- Assume term independence (i.e., BoW)
- Assume query and document to be the same
- Lack of “predictive adequacy”
 - Arbitrary term weighting
 - Arbitrary similarity measure
- Lots of parameter tuning!

Probabilistic ranking principle

Basic concepts in probability

- Random experiment
 - An experiment with uncertain outcome (e.g., tossing a coin, picking a word from text)
- Sample space (S)
 - All possible outcomes of an experiment, e.g., tossing 2 fair coins, $S=\{HH, HT, TH, TT\}$
- Event (E)
 - $E \subseteq S$, E happens iff outcome is in S, e.g., $E=\{HH\}$ (all heads), $E=\{HH, TT\}$ (same face)
 - Impossible event ($\{\}$), certain event (S)
- Probability of event
 - $0 \leq P(E) \leq 1$

Basic concepts in probability

- Probability of events
 - Mutually exclusive events
 - $P(A \cup B) = P(A) + P(B)$
 - General events
 - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 - Independent events
 - $P(A \cap B) = P(A)P(B)$

Basic concepts in probability

- Conditional probability
 - $P(B|A) = P(A, B)/P(A)$
 - Bayes' Rule: $P(B|A) = P(A|B)P(B)/P(A)$
 - For independent events, $P(B|A) = P(B)$
- Total probability
 - If $A_1, \dots, A_n$ form a non-overlapping partition of S
 - $P(B \cap S) = P(B \cap A_1) + \dots + P(B \cap A_n)$
 - $P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B|A_1)P(A_1) + \dots + P(B|A_n)P(A_n)} \propto P(B|A_i)P(A_i)$
 - This allows us to compute $P(A_i|B)$ based on $P(B|A_i)$

Theoretical justification of ranking

- As stated by William Cooper

“If a reference retrieval system’s response to each request is a ranking of the documents in the collections in order of decreasing probability of usefulness to the user who submitted the request, where the probabilities are estimated as accurately as possible on the basis of whatever data made available to the system for this purpose, then the overall effectiveness of the system to its users will be the best that is obtainable on the basis of that data.”

- Rank by probability of relevance leads to the optimal retrieval effectiveness

Justification

- From decision theory
 - Two types of loss
 - Loss(retrieved|non-relevant)= a_1
 - Loss(not retrieved|relevant)= a_2
 - $\phi(d_i, q)$: probability of d_i being relevant to q
 - Expected loss regarding to the decision of including d_i in the final results
 - Retrieve: $(1 - \phi(d_i, q))a_1$
 - Not retrieve: $\phi(d_i, q)a_2$

Justification

- From decision theory
 - We make decision by
 - If $(1 - \phi(d_i, q))a_1 < \phi(d_i, q)a_2$, retrieve d_i
 - Otherwise, not retrieve d_i
 - Check if $\phi(d_i, q) > \frac{a_1}{a_1 + a_2}$
 - Rank documents by descending order of $\phi(d_i, q)$ would minimize the loss
 - Quiz: Can you prove it?

According to PRP, what we need is

- A relevance measure function $F(q,d)$
 - For all q, d_1, d_2 ,
 $F(q,d_1) > F(q,d_2)$ iff. $p(\text{Rel} | q,d_1) > p(\text{Rel} | q,d_2)$
 - Assumptions
 - Independent relevance
 - Independent loss
 - Sequential browsing
- Most existing research on IR models so far has fallen into this line of thinking... (Limitations?)

Summary

- Boolean model
- Vector space model
 - Basic idea
 - TF-IDF
 - Similarity measure: Euclidean distance vs cosine similarity
- Probabilistic ranking principle
 - Essential concepts in probability
 - Justification of ranking by relevance

Readings

- [MRS] CH 6, CH 11.